

AI-Native Software Engineer – 24 Hour Assignment

Objective

Build a lightweight AI-powered Website Audit Tool that:

1. Accepts a single URL
2. Extracts and displays key factual metrics
3. Uses AI to generate structured insights and recommendations

The goal is to evaluate how you structure AI-driven systems, not to build a production-ready product.

Context

EIGHT25MEDIA builds high-performing marketing websites focused on SEO, conversion optimization, content clarity, and UX.

We are exploring internal AI tools that help us quickly evaluate webpages and identify improvement opportunities.

Scope

- Analyze a **single page only**
 - No multi-page crawling required
 - Keep the solution focused and well-scoped
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Requirements

1. Factual Metrics (Required)

Your tool must extract and clearly display key metrics such as:

- Total word count

- Heading counts (H1–H3)
- Number of CTAs (buttons or primary action links)
- Number of internal vs external links
- Number of images
- % of images missing alt text
- Meta title and meta description

These metrics must be clearly separated from AI-generated insights.

2. AI Insights (Required)

Using the extracted data and page content, generate a structured analysis covering:

- SEO structure
- Messaging clarity
- CTA usage
- Content depth
- Obvious UX or structural concerns

Insights must:

- Be grounded in the extracted metrics
 - Be specific and non-generic
 - Clearly reference the factual data
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3. Recommendations (Required)

Provide:

- 3–5 prioritized recommendations
- Clear reasoning tied to extracted metrics

Recommendations should be actionable and concise.

Interface

Provide one of the following:

- Deployed web app link

- Local web app with clear setup instructions
 - CLI tool
 - API endpoint returning structured output
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Prompt Logs (Required Deliverable)

Include prompt logs or reasoning traces that show:

- The system prompt(s) used
- The user prompt(s) were constructed
- The structured inputs sent to the model
- The raw model outputs (before formatting, if applicable)

We want visibility into how you structure and orchestrate the AI layer.

You may redact API keys or sensitive information.

Technical Expectations

Use any stack you prefer.

We are evaluating:

- Clean separation between scraping and AI analysis
- Structured outputs
- Prompt design quality
- Engineering clarity
- Thoughtful trade-offs

Keep the solution simple and practical.

Deliverables

1. GitHub repository
2. Deployed link or runnable instructions
3. README including:

- Architecture overview
 - AI design decisions
 - Trade-offs
 - What would you improve with more time
4. Prompt logs/reasoning traces
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Evaluation Criteria

We are assessing:

- AI-native thinking (not just API usage)
- Ability to ground AI in structured data
- Code quality and organization
- Quality and usefulness of insights
- Practical relevance to a web agency